

Sub Code:	CS/IS/AI/AD/CY/CI41	Sub:	Statistics, Probability and Linear Programming			Test:	02
Time :	3: 00 pm To 04.00 pm	Term:	03:03:2025 To 21-06-2025			Marks:	30
Date:	16.06.2025	Semester:	IV	Section:	All		

Note: Answer any **TWO** full questions. Each question carries 15 marks

Q. No.		Questions	Blooms Level	CO's	Marks									
1.	(a)	Define Absorbing state	L1	CO3	2									
	(b)	Write a short note on Type I and Type II errors	L2	CO4	3									
	(c)	Solve using graphical method, $\min z = 200x + 500y$, subject to the constraints $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.	L5	CO5	5									
	(d)	A supermarket has a single cashier. During peak hours, customers arrive at a rate of 20 per hour. The average number of customers that can be processed by the cashier is 24 per hour. Calculate i)The average number of customers in the queuing system ii)The average time a customer spends in the system iii)The average number of customers in the queue	L4	CO3	5									
2.	(a)	Write the expression to determine the expected waiting time of a customer in the M/M/1 system	L1	CO3	2									
	(b)	A manufacturing company makes two types of television sets; one is black and white and the other is colour. The company has resources to make at most 300 sets a week. It takes Rs 1800 to make a black and white set and Rs 2700 to make a colored set. The company can spend no more than Rs 648000 a week to make television sets. If it makes a profit of Rs 510 per black and white set and Rs 675 per colored set, how many sets of each type should be produced so that the company has a maximum profit? Formulate this problem as a LPP given that the objective is to maximize the profit.	L2	CO5	3									
	(c)	There are 2 white marbles in bag A and 3 red marbles in bag B. At each step of the process a marble is selected at random from each bag and the two marbles selected are interchanged then find (i) the transition probability matrix (ii) what is the probability that there are 2 red marbles in A after 3 steps	L3	CO3	5									
	(d)	A fertilizer mixing machine is set to give 12 kg of nitrate for quintal bag of fertilizer. Ten 100 kg bags are examined, the percentage of nitrate per bag are as follows: 11, 14, 13, 12, 13, 12, 13, 14, 11, 12. Are there any reasons to believe that the machine is defective?	L4	CO4	5									
3.	(a)	Is the stochastic matrix $\begin{bmatrix} \frac{1}{2} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \end{bmatrix}$ regular? Justify.	L1	CO3	2									
	(b)	Find Minimum $z = 5x + 4y$ subject to the constraints $x + 2y \leq 10$, $x + y \geq 8$, $2x + y \geq 12$, $x \geq 0$, $y \geq 0$ by graphical method.	L2	CO5	3									
	(c)	Patients arrive at a doctor's clinic according to Poisson distribution at a rate of 30 patients per hour. The waiting room does not accommodate more than 9 patients. Examination time per patient is exponential with mean rate of 20 per hour. Find the: i) Probability that an arriving patient will not wait ii) Average number of patients in the clinic iii) Expected time a patient spends in the clinic	L5	CO3	5									
	(d)	The following table shows the result of an experiment to investigate the effect of vaccination induced on the animals against a particular disease. Use χ^2 - test to test the hypothesis that there is no difference between vaccinated and unvaccinated groups i.e. vaccination and this disease are independent	L4	CO4	5									
		<table><tr><td></td><td>Got disease</td><td>Did not get disease</td></tr><tr><td>vaccinated</td><td>9</td><td>42</td></tr><tr><td>Not vaccinated</td><td>17</td><td>28</td></tr></table>		Got disease	Did not get disease	vaccinated	9	42	Not vaccinated	17	28			
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